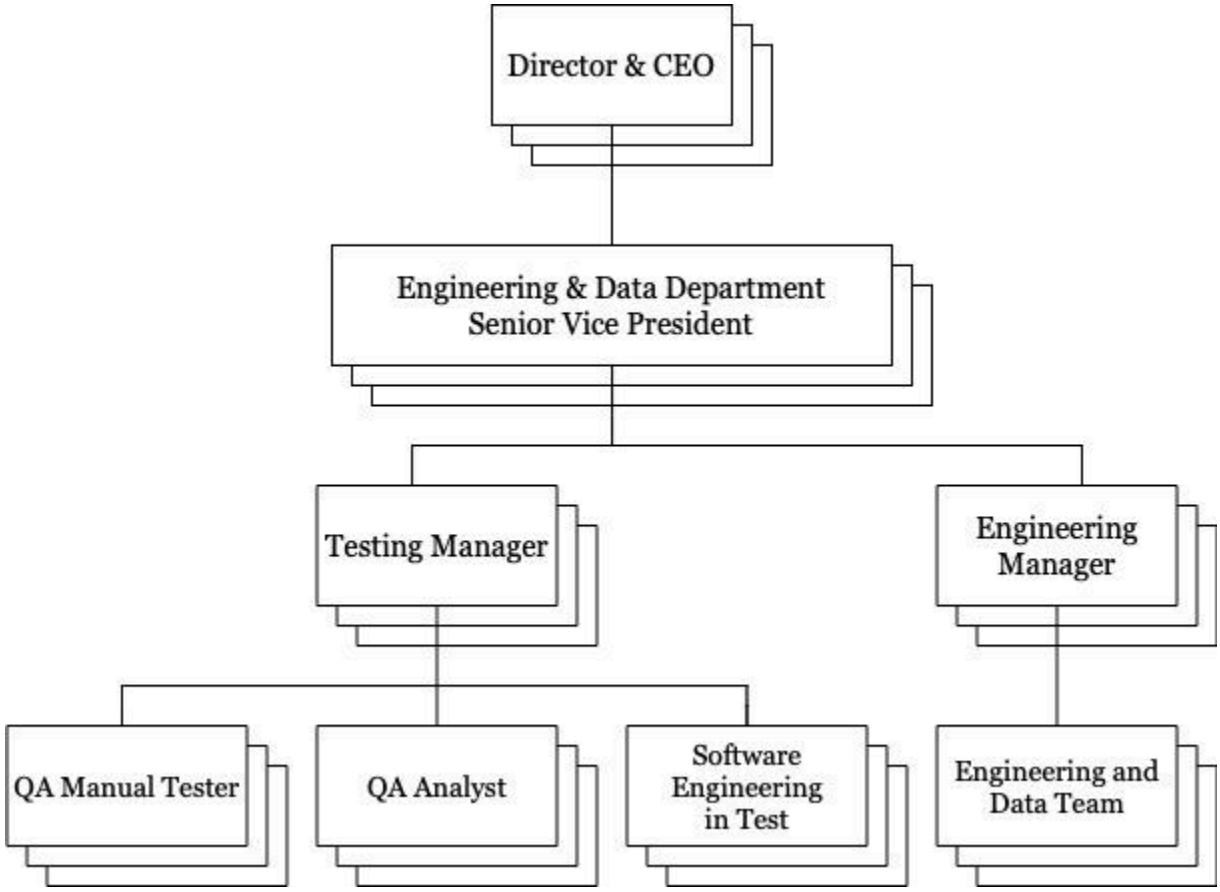
Group 2:

- Madeline Elena - 2502012552
- Noor Azizah - 2501998302
- Rara Atsilah - 2502051686

Testing Structure in Ruangguru



Testing Manager:

- Coordinating testing activities
- Assessing and managing risks
- Making recommendations
- Overseeing planning of systems and tests

Quality Assurance Manual Tester:

- Identify, record and track bugs/defects in JIRA
- Doing exploratory testing to catch possible negative or edge cases
- Maintaining and executing the manual tests that will increase the quality of software products

Quality Assurance Analyst:

- Drive product/feature quality, assess risk and ensure feature delivery to high quality standards
- Plan, develop, and execute testing strategies across Ruangguru products and platforms
- Create a test plan
- Run the test plan (run manual tests for test cases not found or test cases that fail on automation tests)
- Maintenance of test cases (including adding or updating test cases)
- Generate the test plan result
- Reporting bugs
- Define quality metrics and apply measurements to determine test effectiveness, test efficiency, and measure the overall quality of the product/project.
- Leading quality assurance efforts aligned with product/project goals

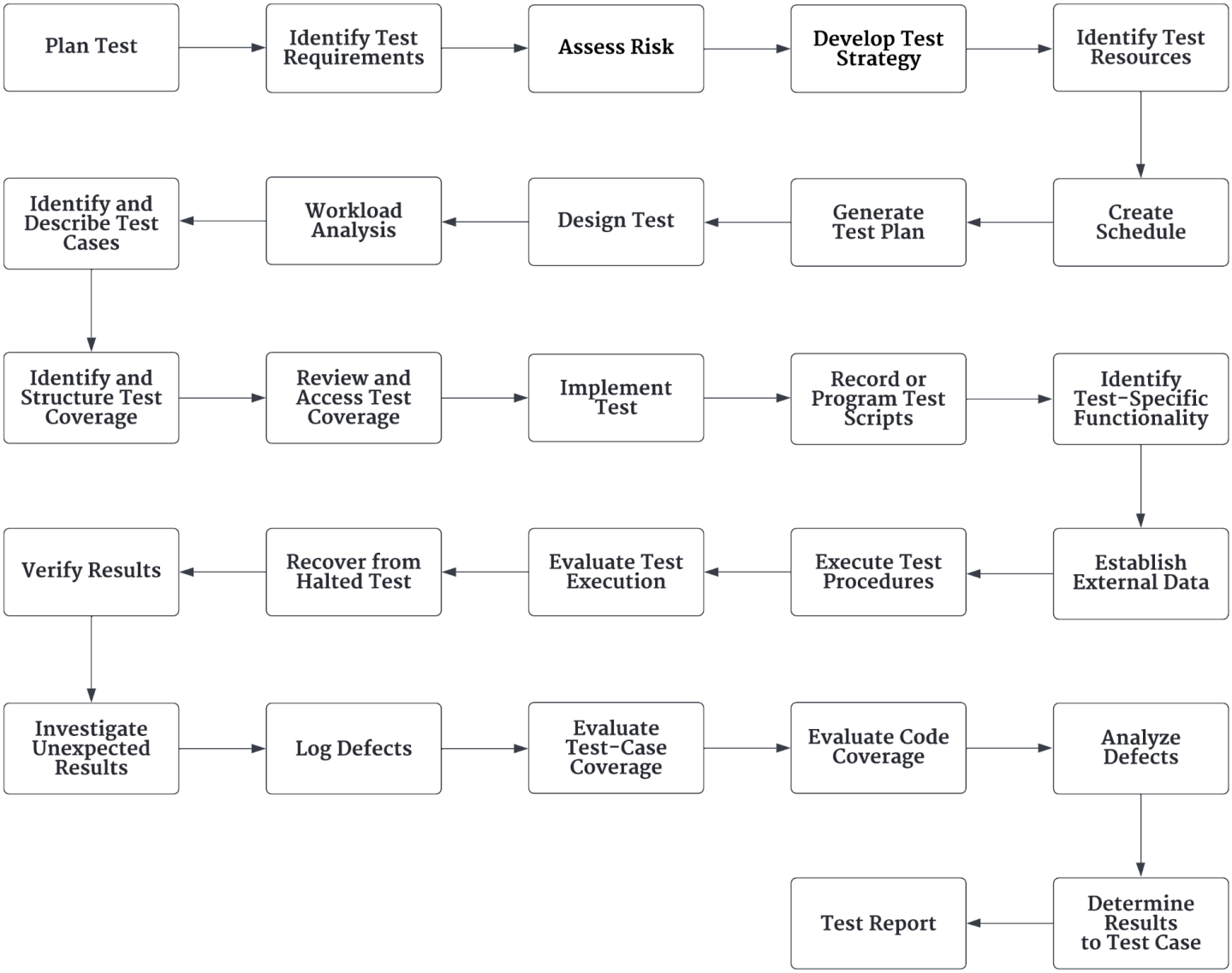
Software Engineering in Test:

- Develop a test strategy
- Analysis of user stories to define test scenarios and test cases, which includes story acceptance criteria based on DoD
- Responsible for Automating testing using test frameworks including, API, web, and mobile based on test cases.
- Responsible for monitoring the development and use of products at all levels with the aim of improving product quality.
- Create test harnesses and necessary infrastructure.

Engineering and Data Team:

1. Data Engineer
2. Data Scientist
3. Data Analyst
4. Tech Ops
5. Data Warehouse Engineer
6. Fullstack Engineer: Analytics
7. Android Engineer
8. Infrastructure Engineering
9. Backend Engineer
10. Frontend Engineer
11. iOS Engineer
12. Fullstack Engineer
13. IT Consultant
14. Engineering Supervisor

Testing Process in Ruangguru



Marketplace Test Plan Development

LA17 Group 2:

1) Madeline Elena - 2502012552
2) Noor Azizah - 2501998302
3) Rara Atsilah - 2502051686

Case Study:

A marketplace site has features for both seller and buyer.

The Objective of the Test:

- We do testing to evaluate the quality of a software product and also to improve it by identifying defects. Software testing strives to achieve its objectives but has certain limitations. However, adherence to the established objectives ensures effective testing.
- To reduce the possibility of errors in software or applications

The Risk of the Test:

- The risks that may occur when conducting a test plan are, lack of availability of required hardware, software and data, late errors. There are also indirect risks such as excessive delays in repairing defects found in testing or problems with getting professional system administration support for the test environment.

The Scope of the Test for Seller:

Items	Features	Procedure to be Tested
Add Products	Product Image	1. Insert product photos/video directly with camera or choose from gallery 2. Delete photo/video 3. Test image quality 4. Image successfully inserted 5. Min. 1 product image 6. Max. 10 product images 7. Move order of images
	Product Name	1. Input product name 2. Min. 5 characters letters 3. Max. 255 character letters
	Product Description	1. Input product description 2. Min. 20 characters letters 3. Max. 3000 character letters

	Categories	1. Select product category based on all product categories view 2. Last used category display
	Dangerous Products	1. Selecting dangerous product settings by selecting contains batteries/magnets/liquids/combustible materials or the product doesn't contain harmful ingredients
	Variation (e.g. Type, Color, and Size)	1. Input product variations 2. Insert a photo according to the variation inputted 3. Set variation info based on price and stock 4. Test editing in bulk 5. Save changes or discard changes
	Price	1. Set price 2. Price limit = Rp99.00 - Rp150000000.00
	Stock	1. Set stock quantity 2. Min. 1 stock
	Wholesale Price (Multiple Price)	1. Input the minimum - maximum order 2. Input unit price 3. Added new wholesale price 4. Save changes or discard changes
	Delivery Estimate (Weight and Size)	1. Set product weight 2. Add package size (Width, length and height by cm) 3. Choose the expedition service that can be used 4. Min. weight 1g 5. Max. weight 50000g 6. Save changes or discard changes
	Product Condition (New or Preloved)	1. Determine the condition of the product by choosing the new or used option 2. Confirm product condition
	Pre-Order Feature	1. Activate the pre-order button when the product takes more time to ship (e.g. the product must be made first)

	Share on Social Media (Promotion)	1. Testing the success of the product sharing feature on social media
	Save Changes or Discard Changes	1. Ensure the product has finished saving changes 2. Ensuring unadded products can be discarded changes
	Show Product on Shop	1. Ensure the product is successfully displayed in the shop
Process Transaction from the Buyer	Incoming Order	1. Check notifications from incoming order 2. Ensure the customer order match with the seller"s sales order
	Product Needs to Ship	1. Testing the need to be shipped product listing by date/order 2. Ensure there are no order missed or unlisted
	Product Sent	1. Testing the shipping data record 2. Match the product with the order
	Product Received	1. Notification from the customer about the arrived product
	Funds Released	1. Ensure the user had already received the product
	Withdrawal of Funds	1. Ensure the product's order cancellation confirmation

The Scope of the Test for Buyer:

Items	Features	Procedure to be Tested
Browse the Products	Search Column	1. Columns can be filled with characters or numbers 2. Can search for products by scanning QR code or inserting image 3. Relevant search results 4. User search history available 5. Clear search history available 6. Popular search history available 7. Cancel button to cancel the product search

	Sort Products (Related, Newest, Best Seller, Price)	1. Use the sort feature to sort the content according to the desired order (e.g. sort by latest product) in ascending or descending order 2. Sorting feature can work accurately and properly arranged
	Search Filters	1. Ensure the search filters show: a. Payment method (COD or installment) b. Product location c. Category d. Price range (Min-max) e. Product color f. Seller type (Official store, overseas seller, and unlabeled seller) g. Rating (5 stars, 4 stars, 3 stars, 2 stars, and 1 stars) h. Shipping options (Regular, thrifty, cargo, same day, instant, next day, onsite pick up) i. Promotional programs (Free shipping, cashback, discounts, wholesale) j. Brand 2. Reset button or use search filter 3. Ensure users can select the search filter they want to use
Choose the Products	Product Image	1. Ensure product images are clear and match the product name and description 2. Ensure product images can be swiped and tapped to see the full view 3. Back button is available to return to the previous view
	Product Name	1. Ensure product name match the product image and description 2. Ensure product name appears on product display
	Seller Type	1. Ensure the seller type appears on the product display for easy

		identification
	Product Details	1. Ensure the product details show: a. Stock b. Guarantee period c. Warranty type d. Origin of shipping e. Brand f. Product description 2. Ensure product description match the product name and image
	Price	1. Ensure the price output has appeared 2. Users can see discount prices when sellers are on promotion 3. Users can see product prices before and after discounts
	Rating & Comments	1. Ensure users can see product ratings and reviews section 2. Ensure users can see buyer ratings and reviews section 3. User can see number of product reviews 4. Users can choose to display ratings based on all ratings options, comments only, buyer photos/videos, comments from the highest rating/lowest rating 5. Users can filter reviews by product variety (e.g. Product color and size) 6. Apply variations button to display rating and comments by variation
	Sold Amount	1. Ensure the sold amount is the same as the purchase amount
	Shop Vouchers	1. Tap the shop voucher section 2. Claim the available shop vouchers 3. Tap use button to use voucher 4. Tap the close button to close the voucher page and return to the product overview
	Available Payment Method	1. Make sure the payment method info match with the seller payment

		option partnership
	Shipping Information (Shipping Location & Available Shipping Options)	1. Show the seller's shop location in real life for shipping fee estimation 2. Show available shipping courier option that the seller cooperate with
	Product Variations	1. Make sure product variety selection options are available 2. Select one of the product's variations 3. Make sure the add/remove item feature is available 4. User can add the selected variety product to cart or buy now
	Product Quantity	1. Ensure customers can see the amount of quantity available 2. Ensure the amount of stock available in accordance with the amount that is still available
	Visit the Shop	1. Ensure the page will directed to the seller's profile shop page
	Seller Chat	1. Ensure the connection between the seller and buyer doesn't disconnect through system error/malfunction 2. Users can make offers on selected products via the chat menu 3. Quick message available 4. User can send image from gallery or photo with camera 5. Users can submit products from the seller's shop for discussion 6. User can submit order list to remind seller for delivery 7. Sticker feature is available for friendly chats with the seller
	Add to Cart	1. Users can add products to cart if they don't want to buy now 2. Users can choose the desired product variation if available 3. User inputs product quantity 4. User taps add to cart button

		5. Check if the product is already in shopping cart
Pay for the Product	Shipping Address	1. Users can input the shipping address as well as the corresponding pin layout 2. Users can choose an address that has been inputted before
	Added New Shipping Address	1. Add new address button 2. Ensure that user can input: a. Recipient's contact (Full name and phone number) b. Recipient address (Province, city, sub-district, zip code, street name, house number, other details) c. Set recipient address as (Home or office) 3. Save changes or discard changes
	Product Overview	1. Ensure product summary view appears in detail (Shop name, product name, selected variety, product price, quantity) 2. Ensure the buyer's order match with the seller's product description data and the billing detail
	Shipping Options	1. Ensure shipping service options are available 2. Users can choose orders to be delivered directly to their homes or pick up on the spot (With agents, drop points, or the nearest partnership minimarket) 3. Ensure the option provide for the customer match with the seller partnership data 4. Shipping option confirmation button
	Add Notes	1. Ensure the note text box was editable and saved until order reaches the seller 2. The note box is optional

	Total Product Quantity and Price Amount	1. Ensure the order quantity is displayed on the screen 2. Ensure the total order appears on the screen
	Voucher Promotion	1. Claim the voucher promotion 2. Users can choose free shipping vouchers and available discount/cashback vouchers 3. Ensure the feature can input promotional voucher codes offered by sellers 4. There is an OK button to approve the use of the voucher
	Payment Methods	1. Check the payment methods 2. Sign up to make a payment 2. Log into the payment accounts/filling in their payment information 3. Choose one of the available payment methods 4. Make sure the payment method feature displays options: a. COD b. Bank transfer c. Credit/debit card d. Credit card installments e. eWallet (e.g. Dana and ShopeePay) f. Minimarket partnership (e.g. Indomaret) g. PayLater 5. Showing handling fees for COD payment options 6. Payment method confirmation button
	Payment Detail	1. Ensure payment details show details: a. Product subtotal b. Subtotal delivery c. Handling fee d. Total payment 2. Check subtotal of the payment and will be directed to the payment method page for transaction process

		3. Ensure the price calculation are correct and consist of tax, shipping fee, product's price, and applied discounts/vouchers
	Order Payment	1. Tap the create order button to proceed with the payment 2. Seller sends invoice 3. User can pay the bill given 4. Ensure the Payment Verification steps already done and approved by the system

Entry, Continuation, and Exit Criteria for User Acceptance Test:

- Entry
  - Availability of bug tracking tools
  - All planned fixes and software's final versions have been implemented
  - All side tests and main system test have been run through and the result had already been fixed if there are any errors
- Continuation
  - All the software has reach the requirement and agreed to be executed
  - No other changes had been made after the finalization process
  - Weekly meetings for strict system quality control to monitor the performance
- Exit Criteria
  - All the mandatory tagged error has been fixed and tested again
  - The bugs are under control and no major interfering bugs founded
  - No crashing/sudden freezing happening in the system
  - The test team had already finished their all testing procedure and fixed any failures

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Review Plan

- **Review Goals**
  - Identification of problems of the components inside the software that needs improvement.
  - Ensuring the components are compatible according to the organizational standards.
  - Communicate developed product phases to the team.
  - Identification of any errors or malfunction in the software components.

- **Preconditions and Items to be Reviewed**
  - **Preconditions for the Review**
    - The duty and responsibility from the reviewed item development team individuals are set.
    - The review to be items are adequately complete for the review process to be useful according to the team leader.
    - Review's objectives statements have been developed.
    - The items' reviews are required in the project plan activity.
  - **Items to be Reviewed**
    - Software Codes
    - Test Plans
    - User Manuals
    - Training Manuals
    - Required Documents
    - Standard Documents

- **Roles, Participants, Team Size, and Time Requirements**
  - **Roles**
    i. Review Leader or Moderator
    Responsible for planning the review, managing the review meeting, and issuing the review report.
    ii. Review Recorder
    Ensure that the time period matches the complexity and size of the item being reviewed. In addition, also responsible for documenting defects, and recording review findings and recommendations.
    iii. Reader
    Responsible for presenting the reviewed item, writing or creating the reviewed item.

iv. Reviewers
Responsible for performing rework on the item under review.
Ensure that the time period matches the complexity and size of the item being reviewed. In addition, also responsible for documenting defects, and recording review findings and recommendations.

- **Participants**
  i. Users/Clients
  The user/client as an outsider has the right to be invited to a review with the aim of obtaining a more impartial evaluation of the item. Of course, the user/client must be present during user manual, requirements, acceptance test plan review, and design to code review.
  ii. Previous Review Team
  It would be prudent for review team members who were involved in the previous and subsequent phases of the lifecycle document under review to participate in this phase. This will be useful in some cases in progress. For example when reviewing design documents, this process will be useful to check correctness, consistency, implementability and traceability are important issues for this review. In addition, these participants can offer different insights and perspectives from group members involved in preparing the document currently under review.
- **Team Size**
  The size of the review team will vary depending on the type, size and complexity of the items being reviewed. Generally, the review team size is between 3 and 7. A team size of at least 3 ensures that reviews will be published.
- **Time Requirements**
  The review session should not be longer than 2 hours and should include a break. Also make sure that the 2 hours allocated are sufficient to review the whole material.

❖ **Training Requirements**

- Review training is training for participants so that the review becomes more effective. The internal technical training staff is responsible for the reviewer training class.
- **Review Training Topics:**
  i. Basic concepts
  ii. Review of quality issues
  iii. Review of standards
  iv. Understanding the material to be reviewed
  v. Defect and problem types
  vi. Communication and meeting management skills
  vii. Review documentation and record keeping
  viii. Special instructions
  ix. Practice review sessions

- **Review Steps**

  There is a standard set of steps that define the review procedures provided for each type of review that the organization wishes to apply. Within each step of the procedure, there are activities and tasks for all reviewing participants. The review plan should also refer to standard procedures if applicable.

  - **Initiation**

    This phase usually uses the checklists to track all the required steps that must be done before the project planning begins.
  - **Preparation**

    After the checklists are complete, the preparation phase begins. The work component examination process by the inspector to detect any defect in the equipment.
  - **Inspection Meeting**

    Readers read work products, part by part and inspectors point out defects for each part.
  - **Reporting Results**

    i. **Defect Database**. A defect is any non-conformance of an item with specified requirements and is a potential cause of performance degradation, as well as failure during the life cycle of a product.

    ii. **Metric Database**. Database metrics help to measure useful quantitative data, such as CPU and storage usage, the number of successful and failed login and database connection attempts, database operations, SQL queries and transactions, and so on. Database metrics can also be used to diagnose and troubleshoot issues with your Database Services resources.
  - **Rework and Follow-Up**

    i. **Rework.** The author makes changes to the work product in accordance with the action plan of the inspection meeting.

    ii. **Follow-Up.** The changes by the author are checked to make sure they are correct.

- **Checklists and other related documents to be disturbed to the participants**
  - **Review Checklist**

    Formally, inspections require the use of a checklist of items that serve as a focal point for review checks and discussions at individual and group levels. The inspector provides a structure and agenda for the review meeting that guides the review activity, identifies focus areas for discussion and evaluation, ensures all relevant items are covered, and helps frame the record keeping and measurement of the review. As a precondition for the development of the checklist, the organization should identify the typical defect types created in previous projects, develop a classification scheme for those defects, and decide on the impact category or severity for the defects. If there is no such flawed data, staff members will need to search literature, industry reports, or organizational archives to retrieve this type of information. Each item that

undergoes review requires a different checklist that addresses specific issues related to quality evaluation for the item.

General Code Review Checklist								
	Developer							
	Name :	General						
	Reviewer							
	Name:	Rara						
No	Problems / Defect Types		Status	Major	Minor	M	I	S
	Design Issues							
1	Does each unit implement a single function?		complete	✓				✓
2	Are there instances where the unit should a single function?		complete	✓				✓
3	Is code cover detailed design		complete	✓				✓
4	Does the code cover detailed design?		complete	✓				✓
	Data items							
5	Is there an input validity check?		Complete	✓				✓
6	Arrays-check array dimensions, boundaries, indices		Pending		✓	✓		
7	Variables - are they all defined, initiated? have correct types and scopes been checked?		Pending		✓	✓		
8	Are all variable used?		Complete		✓	✓		
	Computations							
9	Are there computation using variable with inconsest data types?		Complete	✓				✓
10	Are there mixed-code computation?		Complete	✓				✓
11	Is over- or underflow a possibility (division by zero)?		Complete	✓				✓
12	Are there invalid uses of integers or floating point arithmetic?		Pending		✓	✓		
13	Are there comparisons between floating point numbers?		Complete	✓				✓
14	Are there assumptions about the evaluation order in Boolean expressions?		Pending		✓		✓	
15	Are the comparison operators correct?		Complete	✓		✓		
	Control Flow Issues							
16	Will the program, module, or unit eventually terminate?		Complete	✓		✓		
17	Is there a possibility of an intimate loop, a loop with a premature exit, a loop that never executes?		Complete	✓		✓		
	Interfaces issues							
18	Do the number and attributes of the parameters used by a caller match those of the called routine?		Complete	✓		✓		
19	Does a function or procedures alter a paramaeter is that only meant as an input parameter?		Complete	✓		✓		
20	If there are global variables, do they have corresponding definitions and attributes in all the modules that use them?		Complete	✓		✓		
	Input / output issues							
21	Have all files been opened for use?		Complete	✓		✓		
22	Are all files properly closed at termination?		Complete	✓		✓		
23	It files are declared are their attributes correct?		Complete	✓		✓		
24	Are EOF or I/O errors conditions handed correctly?		Complete	✓		✓		
25	Is I/O butter size and record size compatible?		Complete	✓		✓		
	Portability issues							
26	Is there are an assumed character set, and integer or floating point representation?		Complete	✓		✓		
27	Are their service calls that may need to be modified?		Complete	✓		✓		
	Error Messages							
28	Have all warnings and informational messages been checked and used appropriately?		Complete	✓		✓		
	Comments/Code Documentation?							
29	Has the code been properly documented? are there global, procedure, and line comments where appropriate?		Complete	✓		✓		
30	Is the documentation clear, and correct, and does it support understanding?		Complete	✓		✓		
	Code layout and White Space							
31	Has white space and identation been used to support understanding of code logic and code intent?		Complete	✓		✓		
	Maintenance							
32	Does each module have a single exit point?		Complete	✓		✓		
33	Are the modules easy to change (low coupling and high cohesion)?		Complete	✓				

- **Review Log**
  Log review is a process of review and reporting on a regular basis.

No	Project ID	Project Name	Document Reviewed	Start Date	End Date	Reviewer	Status	Notes
1	PJ001	AA1	Test Plan	15 June	18 June	Azizah	Complete	-
2	PJ002	AA2	Code	15 June	18 June	Azizah	Pending	90% on progress
3	PJ003	AA3	User Manual	16 June	19 June	Madeline	Complete	-
4	PJ004	AA4	Code	17 June	20 June	Madeline	Pending	85% on progress
5	PJ005	AA5	Training Manual	17 June	20 June	Madeline	Complete	-
6	PJ006	AA6	Code	18 June	21 June	Rara	Pending	5% on progress
7	PJ007	AA7	Test Plan	19 June	22 June	Azizah	Complete	-
8	PJ008	AA8	Required Docs	19 June	22 June	Azizah	Complete	-
9	PJ009	AA9	Standard Docs	20 June	23 June	Madeline	Complete	-
10	PJ010	AA10	Test Plan	21 June	24 June	Madeline	Complete	-

- **Summary Report**
  The result from review checklist and review logs, both have completed 97% from the total steps of the procedure. There are neither major defects or errors found in the process. Overall the steps were thoroughly done and successful. But there are still pending status found in some steps and documents and minor bugs tracing that will need further action to be completed.

- **Rework and Follow-Up**
  - **Rework**
    Discuss the defects identified in the preparation phase and the review meeting. The author will try to correct the document based on this flaw and work on it again. The author is responsible for checking for defects and deciding whether or not to rework as not every defect causes rework.
  - **Follow-Up**
    In this phase, the moderator is responsible for ensuring that satisfactory action is taken for all noted defects, process improvements and change requests. Moderators have no obligation to check all rework or corrections in detail, the moderator's main responsibility is to ensure that future analysis data is detailed and correct. So, if the moderator decides that all team members will check the

document and update the same, then the moderator is just taking care of the distribution of roles among the team and collecting feedback from them. The moderator will collect all measurements at each stage of the process to control the review process. For example, the number of defects found, the number of defects found per page, the time spent on the document, the time spent repairing the defects per page, and so on.